Assignment 6

Chinmoy Saikia 20231074
IISER Pune

Question 1

The quantity $llx = 30$. Practice for Periodic Boundary Conditions (PBC):

$$x = \text{modulo}(27.05, llx), \quad y = \text{modulo}(30.05, llx), \quad z = \text{modulo}(-0.03, llx)$$

Answer:
Using the definition of modulo (result always lies in $[0, llx)$):

$$x = \text{modulo}(27.05, 30) = 27.05$$

$$y = \text{modulo}(30.05, 30) = 0.05$$

$$z = \text{modulo}(-0.03, 30) = 29.97$$

$$\boxed{x = 27.05, \quad y = 0.05, \quad z = 29.97}$$

Question 2

Molecular Dynamics Simulation (NVE Ensemble)

A molecular dynamics simulation was performed with $N = 2197$ particles arranged initially on a cubic lattice in a box of size $20 \times 20 \times 20$. The particles interact via the Lennard-Jones (LJ) potential with parameters $\epsilon = 1$, $\sigma = 1$, and a cutoff radius $r_c = 2.5$. The time step used was $\Delta t = 0.005$. Periodic boundary conditions were applied in all three directions.

The simulation was run for 20000 iterations in the microcanonical (NVE) ensemble, i.e., without any thermostat. The equations of motion were integrated using the velocity Verlet algorithm.

Energy Conservation

In the absence of a thermostat, the total energy of the system,

$$E = KE + PE, \tag{1}$$

should be conserved. It was observed that the total energy remains approximately constant throughout the simulation, with only small fluctuations due to numerical integration errors. This confirms that the implementation correctly conserves energy.

Momentum Conservation

The initial velocities were adjusted to ensure zero net momentum. During the simulation, the total momentum of the system remained conserved, indicating correct implementation of pairwise forces obeying Newton's third law.

Kinetic and Potential Energy Behaviour

The kinetic energy (KE) and potential energy (PE) fluctuate around their mean values while exchanging energy with each other. This behaviour is characteristic of the NVE ensemble, where the total energy is fixed but individual components can vary.

From the simulation, it was observed that:

- The kinetic energy fluctuates around $\langle KE \rangle \approx \frac{3}{2} N k_B T$.
- The potential energy fluctuates around a negative value due to attractive interactions between particles.

Typically, for the present simulation (with $k_B T \approx 1$ in reduced units), the energies fluctuate approximately around:

$$\frac{KE}{N} \approx 1.4 - 1.6, \tag{2}$$

$$\frac{PE}{N} \approx -1.5 \text{ to } -2.5, \tag{3}$$

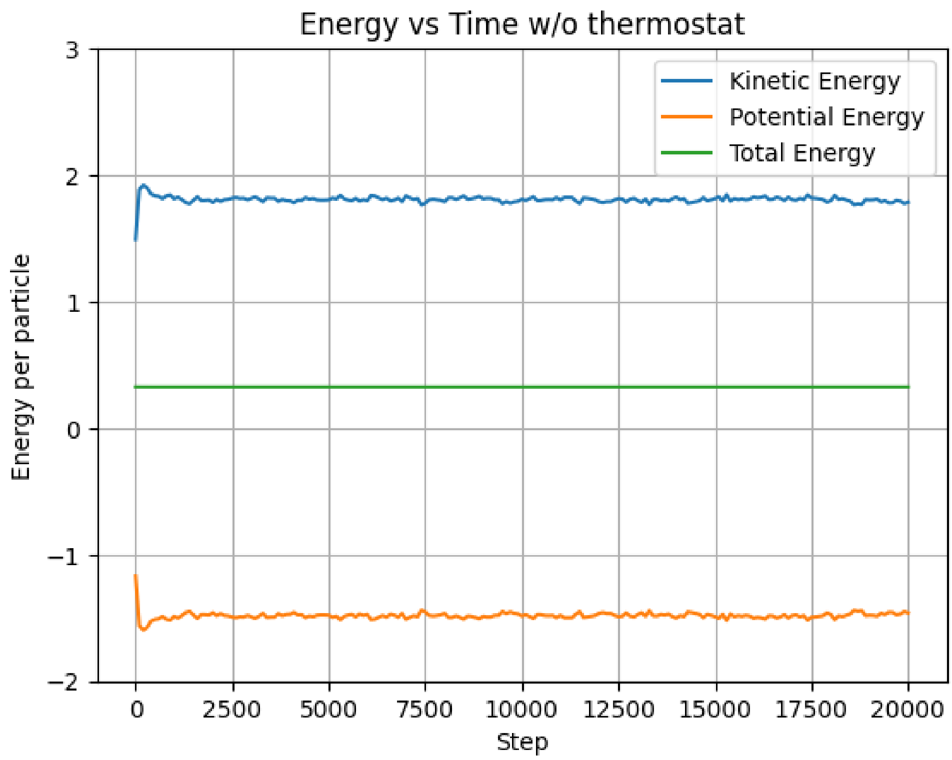
depending on instantaneous configurations.

Conclusion

The simulation successfully demonstrates:

- Conservation of total energy in the NVE ensemble,
- Conservation of total momentum,
- Expected fluctuations of kinetic and potential energies around equilibrium values.

This confirms the correctness of the molecular dynamics implementation.



Question3

Molecular Dynamics Simulation with Thermostat (NVT Ensemble)

A molecular dynamics simulation was performed with $N = 2197$ particles in a cubic box of size $20 \times 20 \times 20$ using periodic boundary conditions. The particles interact via the Lennard-Jones potential with parameters $\epsilon = 1$, $\sigma = 1$, and cutoff radius $r_c = 2.5$. The time step used was $\Delta t = 0.005$.

Unlike the previous case (NVE ensemble), a velocity-rescaling thermostat was applied to maintain the system at a fixed temperature $k_B T = 1$. The simulation was run for 20000 iterations.

Temperature Control

The thermostat rescales particle velocities such that the instantaneous temperature,

$$T = \frac{2}{3Nk_B} KE, \tag{4}$$

remains close to the desired value $T = 1$. As a result, the kinetic energy fluctuates around the target value,

$$\langle KE \rangle = \frac{3}{2} N k_B T. \tag{5}$$

Energy Behaviour

In the presence of a thermostat, the total energy is no longer conserved because the system exchanges energy with an external heat bath. The following behaviour is observed:

- The kinetic energy fluctuates around the value corresponding to $T = 1$.
- The potential energy fluctuates around a negative mean value due to inter-particle interactions.
- The total energy also fluctuates and is **not conserved**, reflecting energy exchange with the thermostat.

From the simulation, the energies typically fluctuate around:

$$\frac{KE}{N} \approx 1.5, \tag{6}$$

$$\frac{PE}{N} \approx -1.5 \text{ to } -2.5, \tag{7}$$

$$\frac{TE}{N} \approx -0.5 \text{ to } 0.2, \tag{8}$$

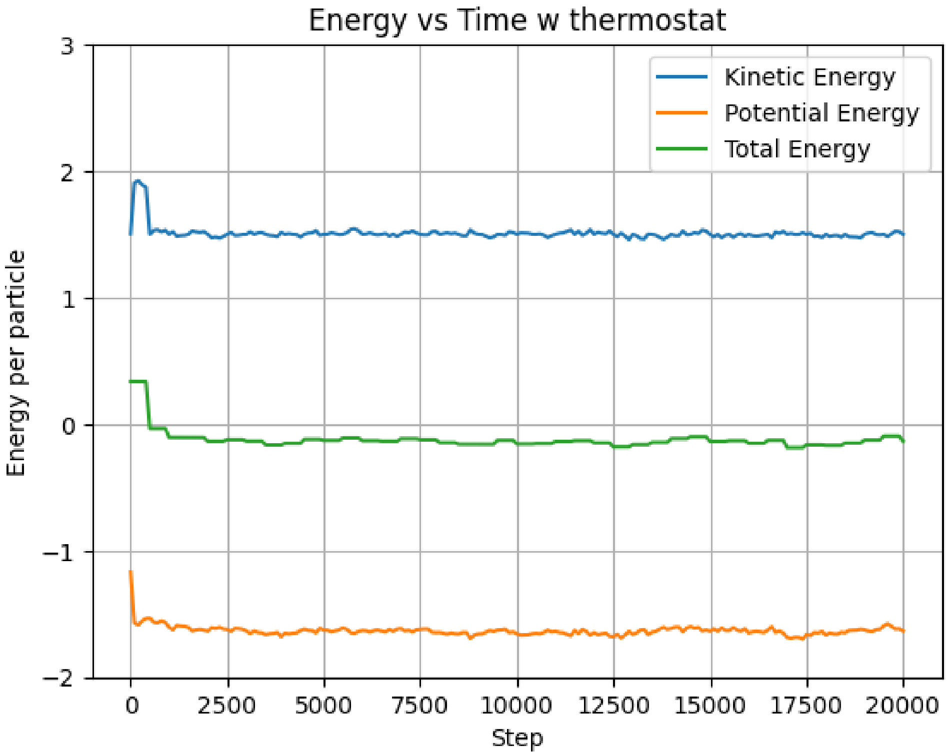
depending on instantaneous configurations.

Conclusion

The simulation demonstrates the expected behaviour of an NVT ensemble:

- Temperature is maintained at the desired value using the thermostat.
- Kinetic energy remains regulated around its equilibrium value.
- Total energy fluctuates due to interaction with the heat bath.

This confirms the correct implementation of the thermostat and temperature control mechanism in the molecular dynamics simulation.



Q4–Q5.

Molecular Dynamics with Neighbour List and Pair Correlation Function

A molecular dynamics simulation was performed with $N = 1200$ particles in a cubic box of size $20 \times 20 \times 20$ using periodic boundary conditions. The particles interact via the Lennard-Jones potential with parameters $\epsilon = 1$, $\sigma = 1$, and cutoff radius $r_c = 2.5$. A time step of $\Delta t = 0.0025$ was used.

Neighbour List Implementation

A Verlet neighbour list was implemented to improve computational efficiency. The neighbour list was constructed using a shell radius

$$r_s = r_c + 2.0 = 4.5\sigma, \tag{9}$$

and updated every 40 iterations.

During the equilibration phase (first 50000 iterations with thermostat), the number of neighbours per particle was monitored. In the interval between 20000 and 30000 iterations, the maximum number of neighbours observed for a particle was:

- Maximum neighbours: 115
- Occurred at step: 24600
- Particle index: 124

For comparison, the expected average number of neighbours within r_s based on uniform density is:

$$\langle n \rangle = \rho \cdot \frac{4}{3}\pi r_s^3 \approx 57.26, \tag{10}$$

where $\rho = N/V$ is the number density. The observed maximum is higher than the mean due to local density fluctuations, which is consistent with a liquid-like system.

Pair Correlation Function $g(r)$

After equilibration, the system was evolved for an additional 250000 iterations in the NVE ensemble. The pair correlation function $g(r)$ was computed by collecting particle pair distances every 100 iterations using a bin size of $\Delta r = 0.1\sigma$.

The first peak of the pair correlation function was found to be:

$$r_{\text{peak}} \approx 1.15\sigma, \quad g(r_{\text{peak}}) \approx 2.39. \tag{11}$$

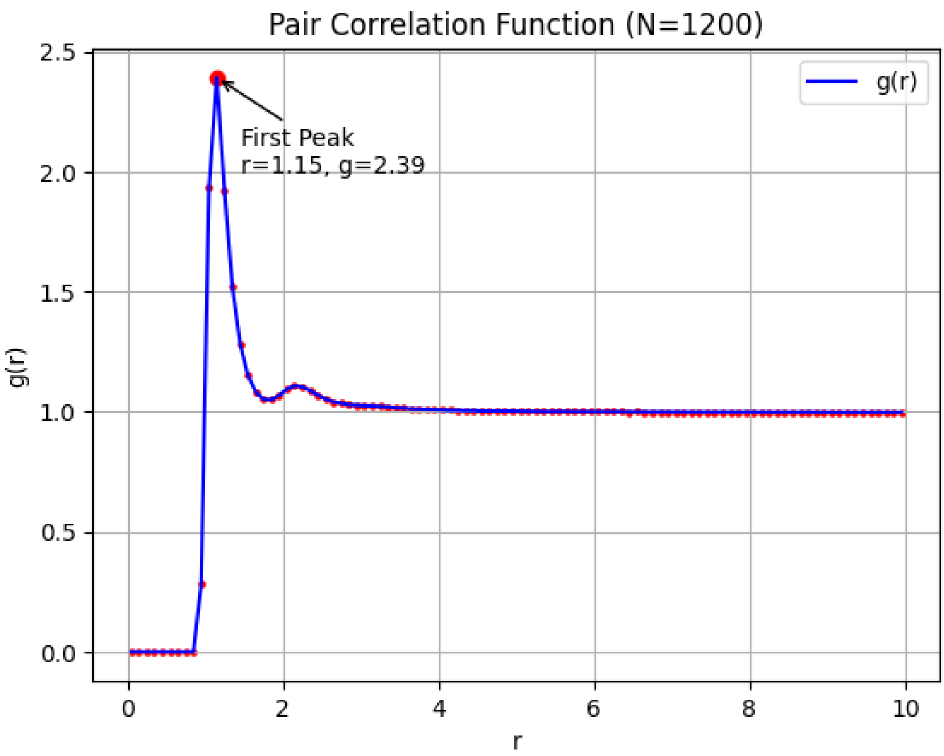
Physical Interpretation

- The first peak at $r \approx 1.15\sigma$ corresponds to the nearest-neighbour distance.
- The peak height $g(r) \approx 2.39$ indicates strong short-range ordering typical of a liquid.
- Beyond the first few peaks, $g(r)$ approaches unity, confirming the absence of long-range order.

Conclusion

The simulation successfully demonstrates:

- Efficient force computation using a Verlet neighbour list,
- Physically consistent neighbour statistics with local fluctuations,
- A well-defined pair correlation function characteristic of a liquid phase.



Question 6

Effect of Increasing Particle Number on Pair Correlation Function

The molecular dynamics simulation was repeated with $N = 2400$ particles in the same box of size $20 \times 20 \times 20$, while keeping all other parameters unchanged. This increases the number density of the system.

The pair correlation function $g(r)$ was computed after equilibration, as in the previous case.

Results

The first and second peaks of the pair correlation function were observed as:

$$\text{First peak:} \quad r \approx 1.15\sigma, \quad g(r) \approx 2.2080, \tag{12}$$
$$\text{Second peak:} \quad r_1 \approx 2.15\sigma, \quad h \approx 1.1206. \tag{13}$$

Interpretation

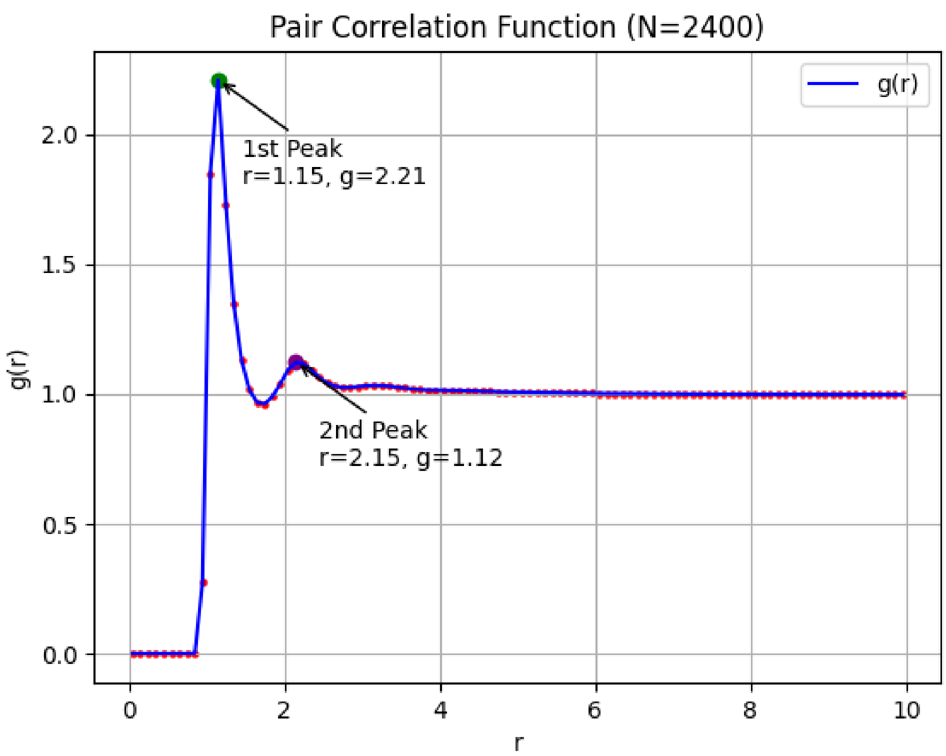
- The position of the first peak remains nearly unchanged at $r \approx 1.15\sigma$, indicating that the nearest-neighbour distance is primarily determined by the Lennard-Jones interaction and is not significantly affected by density.
- The second peak occurs at $r_1 \approx 2.15\sigma$ with height $h \approx 1.1206$, corresponding to the second coordination shell.
- The relatively small height of the second peak indicates weak intermediate-range ordering, consistent with a liquid phase.
- Increasing the number of particles improves statistical sampling, leading to a smoother and more reliable $g(r)$ curve.

Conclusion

The system continues to exhibit liquid-like behaviour, with short-range order and no long-range periodicity. The second peak position and height,

$$r_1 \approx 2.15\sigma, \quad h \approx 1.1206,$$

are consistent with the expected structure of a Lennard-Jones fluid at moderate temperature.



Question 7

Third Peak of Pair Correlation Function for $N = 3600$

The molecular dynamics simulation was performed with $N = 3600$ particles in a cubic box of size $20 \times 20 \times 20$, keeping all other parameters unchanged. The increased number of particles leads to higher density and improved statistical accuracy in the calculation of the pair correlation function $g(r)$.

Results

The peaks of the pair correlation function were observed as:

$$\text{First peak:} \quad r \approx 1.15\sigma, \quad g(r) \approx 2.0255, \tag{14}$$
$$\text{Second peak:} \quad r \approx 2.15\sigma, \quad g(r) \approx 1.0913, \tag{15}$$
$$\text{Third peak:} \quad r_1 \approx 3.25\sigma, \quad g(r) \approx 1.0142. \tag{16}$$

Thus, the position of the (weak) third peak is:

$$r_1 \approx 3.25\sigma.$$

Interpretation

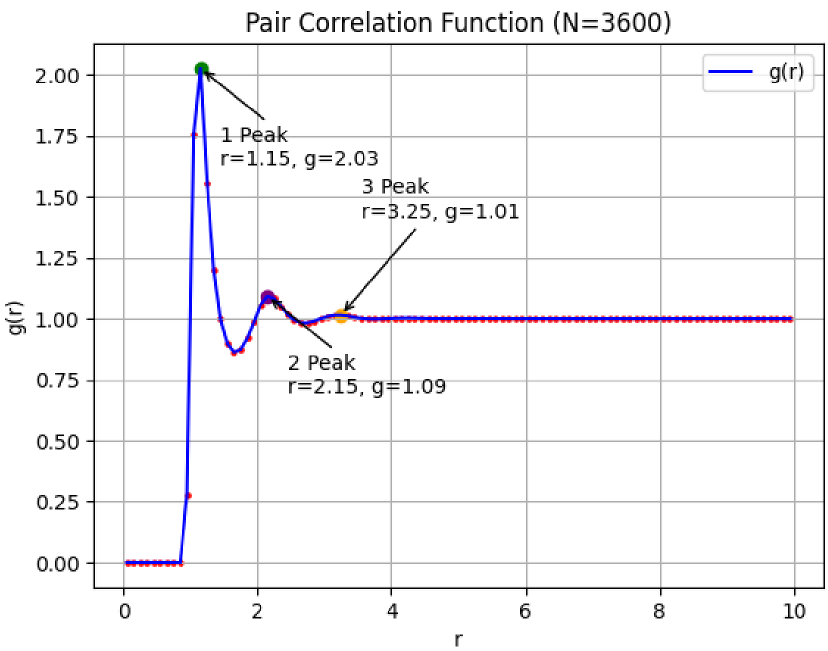
- The third peak corresponds to the third coordination shell of particles.
- Its position at $r \approx 3.25\sigma$ indicates the spatial extent of medium-range ordering in the system.
- The height of the third peak is very close to unity ($g(r) \approx 1.01$), indicating that correlations beyond the second shell are weak.
- This behaviour confirms that the system exhibits liquid-like structure with short-range order and no long-range periodicity.

Conclusion

The third peak position,

$$r_1 \approx 3.25\sigma,$$

is consistent with the expected structure of a Lennard-Jones fluid. The weak nature of this peak further confirms the absence of long-range order, characteristic of the liquid phase.



Question 8

Verification of Maxwell-Boltzmann Speed Distribution

The velocity distribution of particles was analysed during the production phase of the simulation to verify that it follows the Maxwell-Boltzmann (MB) distribution.

The theoretical Maxwell-Boltzmann speed distribution in three dimensions is given by:

$$P(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 \exp\left(-\frac{mv^2}{2k_B T} \right). \tag{17}$$

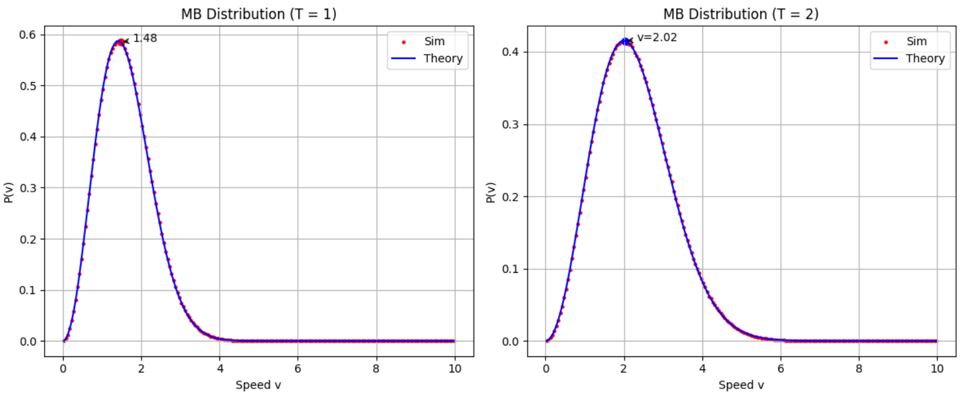
The most probable speed (peak of the distribution) is:

$$v_{mp} = \sqrt{\frac{2k_B T}{m}}. \tag{18}$$

Results

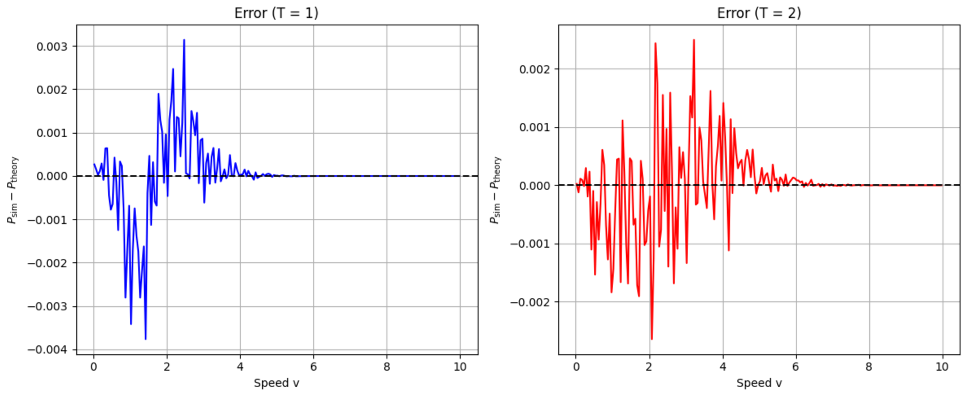
The simulation results were compared with the theoretical distribution for two temperatures:

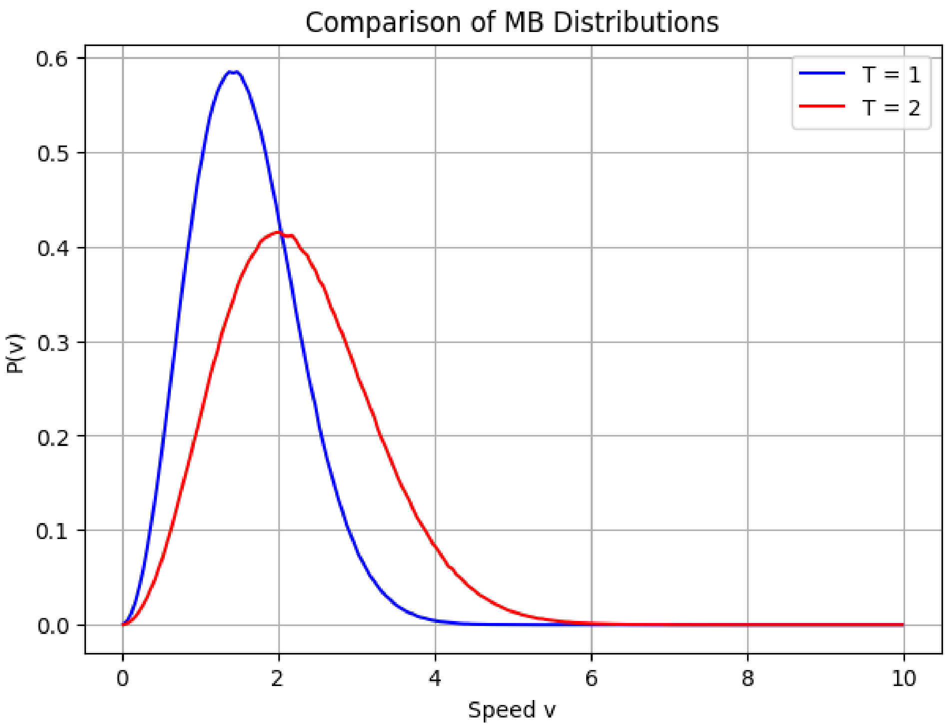
- For $k_B T = 1$:
$$v_{mp}^{\text{sim}} \approx 1.48, \quad v_{mp}^{\text{theory}} = \sqrt{2} \approx 1.414$$
- For $k_B T = 2$:
$$v_{mp}^{\text{sim}} \approx 2.02, \quad v_{mp}^{\text{theory}} = \sqrt{4} = 2.0$$



Discussion

- The simulated distributions closely match the theoretical Maxwell-Boltzmann distribution.
- The peak positions are in excellent agreement with theoretical predictions, with only small deviations due to finite sampling and numerical effects.
- The overall shape of the distribution, including the width and exponential decay at higher velocities, matches the theoretical curve.





Conclusion

The simulation successfully reproduces the Maxwell-Boltzmann speed distribution. The agreement between simulation and theory confirms:

- Correct implementation of velocity initialization and evolution,
- Proper equilibration of the system,
- Validity of statistical mechanics in the simulated system.
